1.125 AGENTIC ENGINEERING / TERM PROJECT PROPOSAL

Kit

A private AI agent for kids, run at home

Questions, privacy, and parent support.

Why do cats purr? **How do you know that?** **恐龙去哪里了?**





Future concept

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A personal AI agent for my daughter

TWO USERS, DIFFERENT NEEDS

Need	My daughter	As a parent
Learning	Ask in Chinese or English	Clear answers with sources
Privacy	Know what gets shared	Control storage and deletion
Help	Ask for a parent with one tap	Receive her help request
Topic summaries	Explore her interests	See topics, not full transcripts
Concerning disclosures	Share if bullied or hurt	Receive an alert under agreed rules

DESIGN REQUIREMENTS

FR1	FR2	FR3	FR4	FR5
Answer at her level	**Keep conversations local**	**Contact a parent**	**Summarize her topics**	**Alert parents to concerns**

The agent can answer, use tools, and notify a parent within agreed rules.

An iPad companion, powered at home

AT HOME



She taps and asks
Chinese or English

Local library + settings

question

answer + source

Mac mini, home AI server



local speech + language models

Agent: the model suggests, rules decide

Reviewed offline library

Notify parent

Daily plan · later

Her lamp · later

PARENT

Help

Help requests
and topic updates

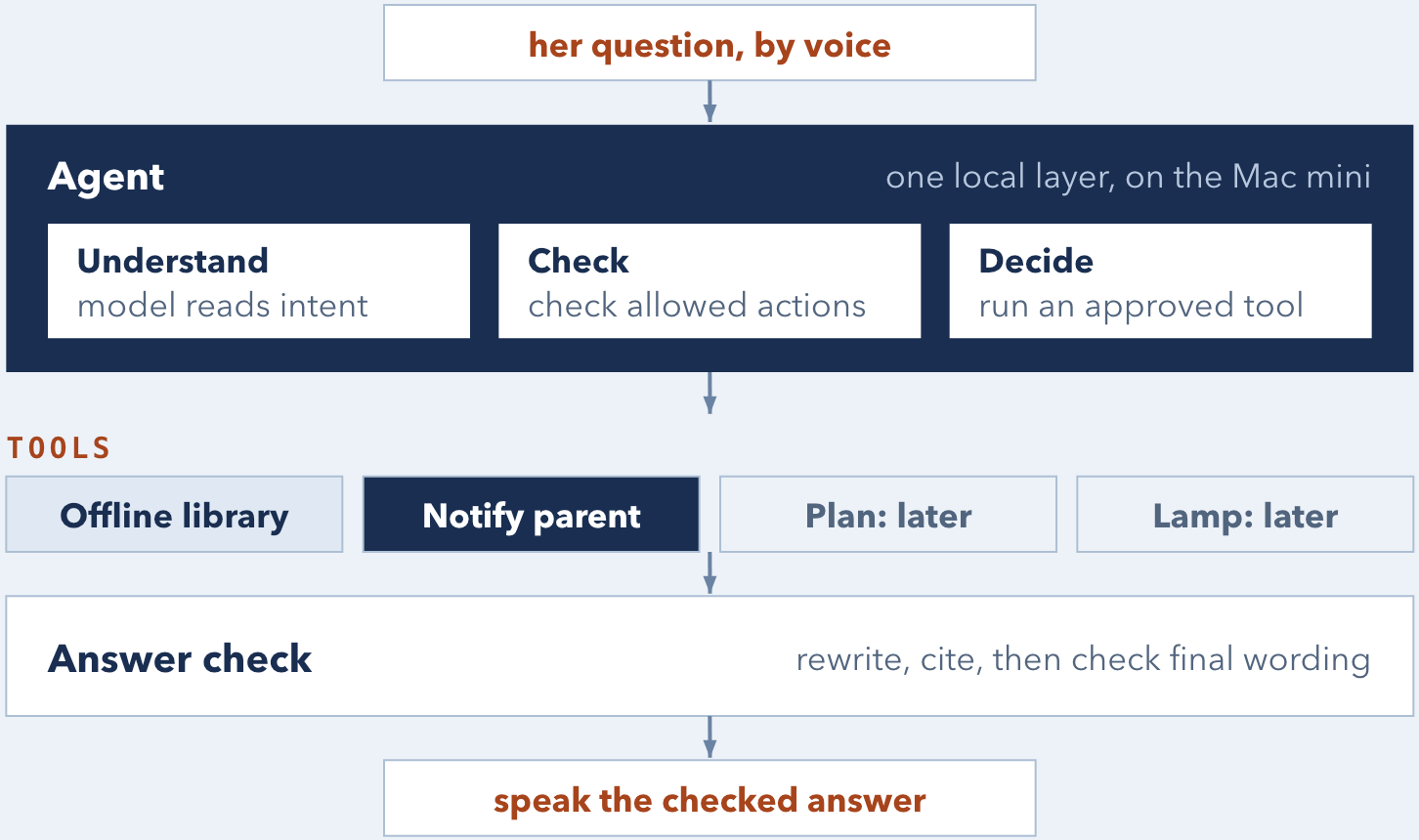
Approved updates

future:
on her wrist



Design: conversations stay local. She controls the mic and can stop at any time.

How Kit handles a question



BUILDING BLOCKS

Local speech recognition
Local language model
Reviewed library in SQLite
Rules and output checks
Authenticated connection

Use reviewed sources.
Say when evidence is missing.

Rules limit actions; tests find failures.

Only approved parent updates may leave the home server.

Help button: sends a request directly, without model approval. Delivery is checked separately.

A small prototype with a clear scope

One complete flow: a spoken question, a sourced answer, and a separate parent help request.

V0.1 PROTOTYPE

- voice Q&A
- iPad cat app
- Wikipedia prototype
- parent button

CORE THIS TERM

- approved tools and inputs
- answer check
- reviewed offline library
- parent alerts + topic summary

LATER

- daily plan
- her lamp
- Apple Watch

RISKS AND TESTS

1. **A small model gets facts wrong**
   Use sources. Admit when evidence is missing.
2. **Missed risks and false alarms**
   Test ordinary and sensitive questions in both languages.
3. **Source coverage is limited**
   Begin with 100 reviewed entries and clear limits.
4. **Real child speech is untested**
   Measure recognition, memory use, and response time.

Preliminary speed: about 3 s on synthetic speech. Child-voice performance is still untested.

How I will test whether Kit works

NOW · OCT 29

voice prototype
requirements and test plan

MIDPOINT · MID–NOV

complete question-to-answer flow
100 reviewed library entries
real child speech measured

FINAL · EARLY DEC

parent topic summaries
supervised trial findings
reproducible demo and tests

VERIFICATION: REQUIREMENTS AND TESTS

Speed

Target: <3 s median to checked speech

Quality

Accuracy, sources, and age fit

Privacy

Check network traffic, access, and deletion

Safety

Test missed risks and false alarms

VALIDATION: CAN SHE USE IT WELL?

Ask → **Explain back** → **Follow up** → **Stop**

Observe with a parent, after an adult trial

Measure real speech first. Report results and failures; revise and re-test.

Demo: ask a question, hear a sourced answer, send a parent request. Publish code and tests.